



European Journal of Educational Research

Volume 14, Issue 1, 73 - 87.

ISSN: 2165-8714

<http://www.eu-jer.com/>

Applying Augmented Reality Technology in STEM Education: A Bibliometrics Analysis in Scopus Database

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Received: July 11, 2024 • Revised: September 9, 2024 • Accepted: October 19, 2024

Abstract: Augmented reality offers diverse potential applications for STEM education, enabling students to engage directly with virtual elements in real-time and providing them with immersive, natural experiences. This study conducted a bibliometric analysis of articles on this topic on the Scopus database to determine some quantitative information, such as general information about publications, countries, institutions, authors with the most contributions, and key trends in applying augmented reality technology in STEM education. An analysis of 201 studies published from 2005 to 2023 using Biblioshiny software and VOSviewer reveals that the United States leads in the number of studies conducted on this issue. Kryvyi Rih National University, Ukraine, has the most studies. The authors who contributed the most studies with the most citations on this issue are Lindner, C. and Rienow, A. from Ruhr University Bochum, Germany. Two primary research trends emerge, focusing on how Augmented Reality technology is utilized, particularly in STEM fields like Chemistry, which combines learning forms with other learning support tools and media such as mobile applications. Secondly, integrating augmented reality and virtual reality technologies into STEM programs at the university level, design of games, and virtual tools. This study offers important data for researchers looking to explore future applications of augmented reality technology within STEM education.

Keywords: *Augmented reality, bibliometrics, Scopus, STEM education.*

To cite this article: Giang, N. T., Dinh, N. V., Ngu, P. N. H., Chau, D. B., Thao, N. P., & Thao, T. T. P. (2025). Applying augmented reality technology in STEM education: A bibliometrics analysis in Scopus database. *European Journal of Educational Research*, 14(1), 73-87. <https://doi.org/10.12973/eu-jer.14.1.73>

Introduction

Scientometrics, or bibliometrics, is a scientific discipline that evaluates research output by quantifying publications. Through this analysis, researchers gain access to quantitative insights into scientific outputs and citation metrics, which assist in guiding their research focus. At the same time, institutions, organizations, and governments can use these insights to allocate research funding effectively (Waltnam & Noyons, 2018). Additionally, university ranking bodies rely on scientometrics as a critical tool for assessing institutional performance (Charlton & Andras, 2007). It also plays a role in evaluating a country's scientific progress and determining its current standing within the global research community (Ha et al., 2020; Pham-Duc et al., 2022).

Recently, educators have focused on developing educational tools powered by new technological advances to address the demands of Industry 4.0 and Society 5.0 (Akturk et al., 2022; Chow et al., 2023). Augmented reality (AR) technology, as an emerging technology, has been rapidly adopted and integrated into educational fields thanks to its diverse capabilities and accessibility to popular devices such as phones and tablets (Bacca et al., 2014). AR refers to bringing virtual objects into physical scenes so that they can be seen, thereby creating a new reality that includes physical objects that we can touch and virtual objects that we can feel and interact with but cannot touch (Assad, 2021). According to Behzadi (2016), AR is an immersive simulation of the real world – in which certain elements are added to the simulated landscape. AR creates an environment where computer-generated information is integrated into a real-life context. AR

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enables the visualization of phenomena that are difficult to see unaided (Wu et al., 2013). AR has demonstrated its versatility at all levels of the educational sector as a suitable step towards technology-based learning environments (Petrov & Atanasova, 2020).

Due to advantages such as enhancing spatial abilities (Lin et al., 2015), combining real and virtual reality (Dunleavy et al., 2009), improving interactions (Kamarainen et al., 2013; Lin et al., 2011; Zarraonandia et al., 2013), reducing the cost of laboratory materials, AR is a precious technology tool for STEM education (Ferrer-Torregrosa et al., 2015). STEM education engages students in engineering design or research, fostering meaningful learning by combining and applying mathematics, technology, and science (Moore & Smith, 2014). STEM education helps improve teachers' teaching or enhance their teaching capacity (Hooker, 2017). STEM education generally helps improve students' academic achievement (Chang et al., 2015), helps connect between-subjects knowledge more easily (Lou et al., 2017), and enhances technological competence (Duran et al., 2014), creativity with accumulated work experiences for students (Lou et al., 2017). In addition, STEM education also helps governments and policymakers focus funding and develop policies on areas that have the best impact on the country (Assefa & Rorissa, 2013).

Determining research trends on using AR in STEM education through bibliometric methods has been of interest to many authors recently. Numerous bibliometric studies on STEM education cover various topics using popular databases like Web of Science (WoS) and Scopus, examining research trends in STEM education from 2004 to 2021 (Zhan et al., 2022), 2015 - 2023 (Kahraman, 2023), 1992 - 2017 (Özkaya, 2019), 1985 - 2021 (Cai et al., 2023), from 1992 - 2022 (Tas & Bolat, 2022), 2006 - 2022 (Phuong et al., 2023), 2017 - 2023 (Aldeia et al., 2023); some other authors conduct STEM education research in certain geographical areas or subjects such as an overview of STEM education research trends in ASEAN in the period from 2000 to 2019 (Ha et al., 2020); Overview of STEM education research at the secondary school level in the period from 2000 to 2020 (Thu et al., 2021); research trends of the engineering design process for STEM education in K-12 in the period of 2011 - 2021 (Ali & Tse, 2023). As for bibliometric research on AR, there have been some studies on this topic on the Scopus and WoS databases. On the WoS database: Arici et al. (2019) researched on AR application trends in science education from 2013-2018; Karakus et al. (2019) compiled AR research trends in education in general, in the period 1999-2018. On the Scopus database, Aldeia et al. (2023), Harnal et al. (2024), and Maulana et al. (2021) presented a bibliometric analysis of some recent applications of AR, highlighting the impacts, challenges, and potential trends of AR in Education over the periods 2006-2020, 2001 - 2020, and 2017 - 2023. Recently, several researchers have shown interest in applying augmented reality (AR) technology in STEM education. However, despite the increasing number of studies on AR in STEM education, there is still no comprehensive overview to guide researchers in identifying gaps and potential research directions. All existing bibliometric studies on this topic have been conducted using the WoS database (Çavdar & Yıldırım, 2023; Talan, 2021; Zhao et al., 2023), with none using the Scopus database—an educational science resource that is considered to be more extensive and diverse than the WoS (Chadegani et al., 2013). This research on AR in STEM education using the Scopus database provides unique insights compared to previous studies. Scopus's broader range of journals and research fields offers a more comprehensive view, uncovering a wider variety of related works. It also helps identify emerging trends and underexplored areas that may not have been captured in the WoS-based studies. With richer data, this study delivers deeper analyses of research collaborations, author networks, and key themes, offering valuable guidance for future research. This makes this research timely and necessary.

We use data from the Scopus database to analyze publications on the use of AR in STEM education. The primary research goals encompass (a) offering an overview of the key characteristics and publication trends, including leading journals and highly prolific researchers; (b) presenting an examination of international collaborations between nations within this area; and (c) identifying prevalent research topics and emerging trends through an analysis of words used in titles, abstracts, and keywords.

Methodology

In this study, the authors utilized a standard scientific mapping process to track the Scopus database, following five key stages: (a) research design; (b) data collection; (c) data analysis; (d) data visualization; and (e) interpretation (Börner et al., 2003; Zupic & Čater, 2015).

The primary research question at the research design stage was: What are the cited references in publications indexed in Scopus related to the application of augmented reality technology in STEM education? The data collection stage includes three sub-stages: data collection, data filtering, and data cleaning.

The first stage is data collection. On December 21, 2023, we conducted a data search in the Scopus database (<http://www.scopus.com>), utilizing advanced search options to input terms and operators in accordance with the platform's syntax. AR technology in STEM education was the focus of the search, with data limited to articles, conference papers, and book chapters. These publications are written in the English language and published before 2024. This study selected the Scopus database due to time constraints and because Scopus is more easily accessible than other databases, such as the WoS.

Query strings on the Scopus database are shown in Figure 1. After the search process, we eliminated some articles lacking information.

```
TITLE-ABS-KEY ("augment* realit*" AND ((educat* OR learn* OR teach*) AND (science* AND technolog* AND engineer* AND mathematic*) OR {STEM})) AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "cp")) AND (LIMIT-TO (LANGUAGE, "English"))
```

Figure 1. Query String in the Scopus Database on Applying Augmented Virtual Reality Technology in STEM Education

The second stage is data filtering. We reviewed titles, abstracts, and keywords and had a comparison and consensus between authors to eliminate publications unrelated or less related to the research issue. Some reasons for eliminating documents are: The search keyword only appears in the background or is only mentioned as an example for comparison, or the research content only mentions one module in STEM. After this step, the remaining documents are 201. The information obtained from these 201 documents was extracted by the authors into CSV format files for further analysis on bibliometric tools, mainly with two software: Bibliometrix and VOSviewer. Microsoft Excel software is also used to synthesize data at different periods. Besides, we continued to use the Scopus database (www.scopus.com) and data from SCIMAGOJR (www.scimagojr.com) to search for information about authors, countries, and journals - including analyzing the quality of these journals through the CiteScore, Scimago Journal & Country Rank (SJR) indexes of 2022.

The third stage is data cleaning. The quality of input data dramatically affects the quality of analysis and research results. Some errors have been corrected by the authors in this step. The most common issues include the same institution having a different name in various papers or the same author presenting their name differently in other publications.

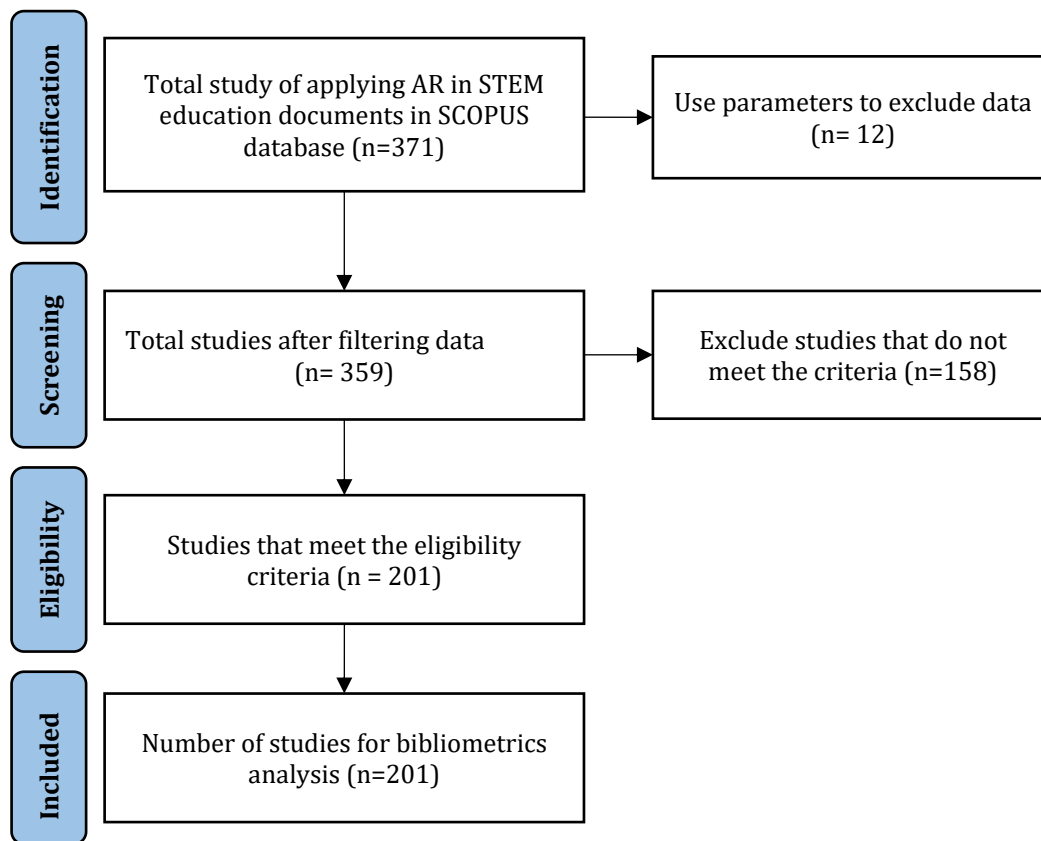


Figure 2. The PRISMA Diagram Describes the Data Filtering Process

During the Data analysis and Data visualization stage, we employed various techniques to extract key insights from the dataset. General details about the dataset, including the annual publication count and cumulative citation numbers, are presented to highlight and explain the research growth trends. Interest in research is demonstrated through information on the contributions of the most productive countries, organizations, journals, and authors. Finally, the data analysis results are interpreted and presented in the Results and Discussion section below.

Results

General Information and Growth Trend

Statistical data indicates that 201 documents will be published across 126 sources, reflecting an annual increase of 21.23% between 2005 and 2023. 682 authors participating in research and publishing on this topic; the average collaboration rate of authors in each document is 4.03, of which the international collaboration rate is 17.91%. In the bibliography, there are 12 articles with only one author (single-authored docs) and 12 authors with only one article (Authors of single-authored docs). There are a total of 507 keywords in this bibliography. The publications in the bibliography cited a total of 7063 references. The average time of publications on this issue is 3.9 years from the time of publication to the time of research conducted. Most of the documents in the bibliography are conference papers (125 out of 201 documents), and the rest are articles (76 out of 201 documents). The documents in the bibliography have an average citation per doc of 13.28, of which 23 documents are cited at least 23 times (h-index=23). Table 1 presents all the information above.

Table 1. Overview of Key Data from the Bibliometric Analysis

Description	Results
Main Information About Data	
Timespan	2005:2023
Sources (Journals, Books, etc.)	126
Documents	201
Annual Growth Rate %	21.23
Document Average Age	3.9
Average citations per doc	13.28
References	7063
Document Contents	
Author's Keywords	507
Authors	
Authors	682
Authors of single-authored docs	12
Authors Collaboration	
Single-authored docs	12
Co-Authors per Doc	4.03
International co-authorships %	17.91
Document Types	
Article	76
Conference paper	125
H-index	23

Figure 3 provides information on the number of publications and cumulative citations by year from 2014 - 2023. The first article about this issue was published in the Scopus database in 2005. However, growth in the number of articles and cumulative citations is negligible (2005, 2009, 2012, and 2013 had one article each year; the remaining years in this period had no articles) from 2005 to 2013. From 2014 to the present, this research topic has received increasing attention from scientists since the number of articles and citations generally increases yearly.

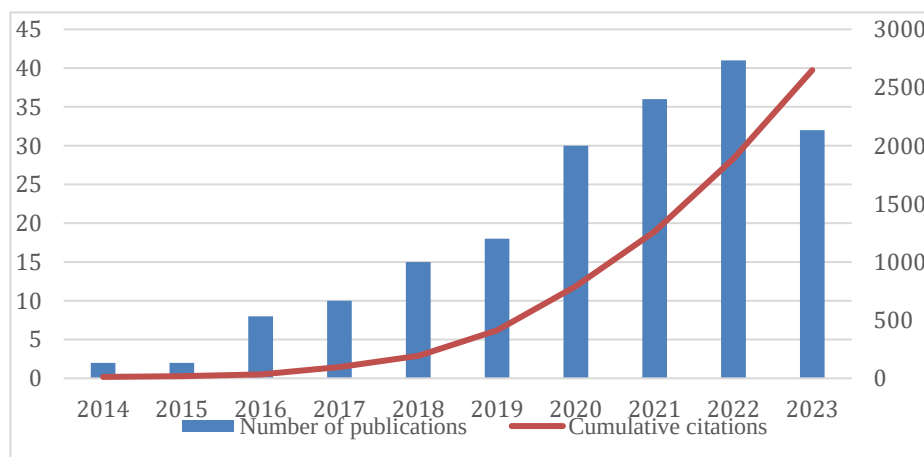


Figure 3. Growth Trends of the Collection's Cumulative Number of Articles and Citations During the Period 2014 - 2023
(Source: Authors' own elaboration)

Contribution by Nations

Authors from fifty-two countries and territories have researched this issue. Table 2 lists the top 10 regions with the highest number of publications on this topic. These countries contributed 162 articles out of a total of 201 articles compiled (accounting for 80.6%). Among these, the United States and European countries (including Germany, Ukraine, Portugal, Greece, Ireland, and the United Kingdom) have more articles than the rest, with a total of 127 articles. The United States alone contributed 59 articles with 409 citations; articles from the UK had the highest total number of citations with 639. Asia has 3 representatives in the top 10, including Malaysia, Indonesia, and Taiwan, with 35 publications.

Table 2. Leading Regions by Number of Publications in the Bibliography

Rank	Country / Territory	TP	TC	TC/TP
1	United States	59	409	6.93
2	Germany	18	124	6.89
3	Malaysia	15	59	3.93
3	Ukraine	15	230	15.33
5	Indonesia	12	45	3.75
6	Portugal	10	73	7.30
7	Greece	9	88	9.78
7	Ireland	9	67	7.44
9	Taiwan	8	99	12.38
10	United Kingdom	7	639	91.29

Notes: TP: Total publications; TC: Total citations; TC/TP: citation per paper

We use VOSviewer to chart academic collaboration on research between countries around the world on this research topic. The criterion for selection is that each country must have at least 3 documents in the bibliography. As a result, 22 countries meet the above criteria, of which 15 countries have academic collaboration with each other (shown in Figure 4). Among these 15 countries, only one representative from Asia, Singapore, in collaboration with United Kingdom scientists, and one from Oceania, Australia. The US has intercontinental collaboration with European countries such as Germany, Switzerland, Italy, and Asia (typically Singapore). German scientists have collaborated with the United States and many other European countries, especially the United Kingdom, Greece, and Austria. The United Kingdom scientists initially collaborated with Finnish scientists, then expanded to other European countries and with Australia.

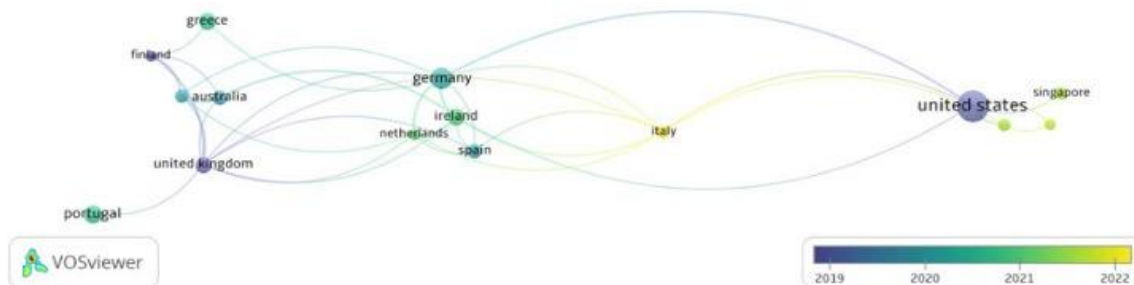


Figure 4. Collaboration Between Countries in the Collection

Contribution by Institutions

Data from Scopus indicate that 160 institutions have been involved in publishing research on the application of AR in STEM education. Table 3 showcases the nine leading institutions responsible for 44 of the 201 publications in this field

Table 3. Top 9 Institutions by Publication Count

Rank	Institution	Country	TP	TC	TC/TP
1	Kryvyi Rih National University	Ukraine	7	144	20.57
2	Ruhr-Universitat Bochum	Germany	6	53	8.83
3	University College Dublin	Ireland	5	34	6.80
4	Purdue University	United States	5	60	12.00
5	University of Central Florida	United States	5	93	18.60
6	Universidade Nova de Lisboa	Portugal	4	5	1.25
7	National Center "Junior Academy of Sciences of Ukraine"	Ukraine	4	67	16.75
8	Centro de Investigação Didática e Tecnologia na Formação de Formadores CIDTFF	Brazil	4	52	13.00
9	Ukrainian State University of Science and Technologies	Ukraine	4	111	27.75

Notes: TP: Total publications; TC: Total citations; TC/TP: citation per paper

Among these nine organizations, organizations from Ukraine contributed the most with 15 articles, distributed to three institutions, the most prominent being Kryvyi Rih National University with seven articles (leading among institutions contributing the most articles in the bibliography); the United States has two universities contributing ten articles.

Contribution by Sources

The studies in the entire bibliography are published in 126 sources. The top 8 most frequently published sources of the collection are shown in Table 4. Among the eight published sources with the highest number of publications in the bibliography, half of the sources belong to the conference proceeding category, three sources belong to the book series category, and there is only one journal.

Table 4. Top 8 Sources with the Most Publications by Frequency

Rank	Source	Type	TP	TC	Scopus Quartile ^a	CiteScore 2022	SJR 2022
1	Ceur Workshop Proceedings	CP	13	185	-	1.1	0.202
2	Lecture Notes in Computer Science	BS	9	38	-	2.2	0.320
3	ASEE Annual Conference and Exposition Conference Proceedings	CP	8	18	-	-	-
4	Journal of Physics: Conference Series	CP	8	48	-	1.0	0.183
5	Communications in Computer and Information Science	BS	6	26	-	1.0	0.194
6	ACM International Conference Proceeding Series	CP	5	6	-	1.1	0.209
7	Education Sciences	JN	4	108	Q2	4.0	0.605
8	Lecture Notes in Networks and Systems	BS	4	1	-	0.7	0.151

^aInformation from Scopus on December 21, 2023

Note: CP: Conference Proceeding; BS: Book Series; JN: Journal; TP: Total publications; TC: Total citations

We see that in the total number of articles published by eight sources in Table 4, there are 57 out of 201 articles in the bibliography (accounting for 28.4%), and the total number of citations of articles in the bibliography published on these sources is 430 out of 2,666 total citations in the bibliography (accounting for 16.1%). The Education Sciences has the largest CiteScore and SJR in the entire collection, and it is the only journal in Table 4's Top 8 list. According to Talan (2021), the Education Sciences is one of the ten journals with the most publications on applying AR technology in STEM education, with publications in the WoS database.

Contribution by Authors

Table 5 presents the 9 most prolific contributors from a total of 682 researchers involved in this field. The largest number of authors are from the United States, with 4 authors, all from Purdue University; Ukraine has 3 authors, all from the National Center of Junior Academy of Sciences of Ukraine; Germany has two authors from Ruhr University Bochum, Lindner, C. and Rienow, A.

Table 5. Top 9 Authors by Publication Count

Rank	Author	Institution/ Country	TP	TC	TC/TP
1	Lindner, C.	Ruhr University Bochum, Germany	6	51	8.5
2	Rienow, A.	Ruhr University Bochum, Germany	6	51	8.5
3	Bilyk, Z.I.	National Center "Junior Academy of Sciences of Ukraine", Ukraine	4	67	16.8
4	Liu, Z.	Purdue University, United States	4	59	14.8
5	Ramani, K.	Purdue University, United States	4	59	14.8
6	Shapovalov, V.B.	National Center of Junior Academy of Science of Ukraine, Ukraine	4	67	16.8
7	Shapovalov, Y.B.	National Center of Junior Academy of Science of Ukraine, Ukraine	4	67	16.8
8	Villanueva, A.	Purdue University, United States	4	59	14.8
9	Zhu, Z.	Purdue University, United States	4	59	14.8

Notes: TP: Total publications; TC: Total citations

The number of publications by the nine authors by year in Table 5 is shown in Figure 5. We used Biblioshiny to produce this figure. The interesting common point of these nine authors is that they are divided into 3 groups: Authors Lindner, C. and Rienow, A. collaborated together on all six articles; the group of authors Liu, Z.; Ramani, K.; Villanueva, A.; Zhu, Z. of Purdue University and the group of authors Bilyk, Z.I.; Shapovalov, V.B.; Shapovalov, Y.B from National Center of Junior Academy of Sciences of Ukraine, Ukraine has a joint collaboration in all their articles in this bibliography. Therefore, the authors in each of these groups have similar statistical indicators in Table 5 and Figure 5. These authors published mainly in the period of 2018 - 2020. From 2020 until now, a group of authors has stopped publishing on this issue.

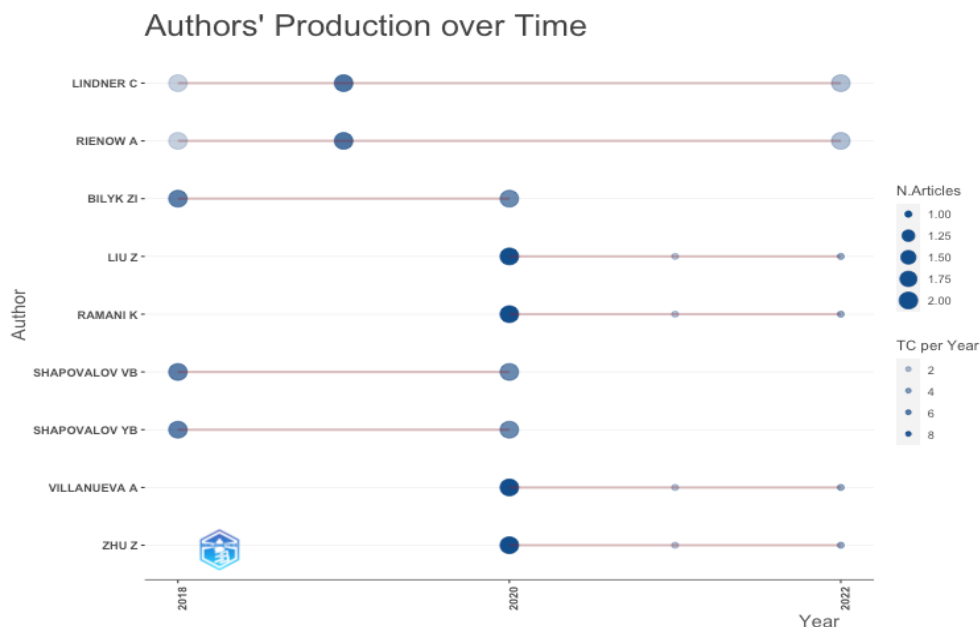


Figure 5. Trends in Publications by Leading Authors (Source: Authors' own elaboration, using Biblioshiny)

The Most Influential Articles

Table 6 presents the Top ten influential articles from the 201 included in the bibliography, ranked by total citation count. Among these ten articles, the one with the most citations is the case study on applying AR in STEM education by the group of authors Jesionkowska J., Wild F., and Deval Y. The article on Meta-AR-App has a number of citations, at least in this group titled An Editing Platform for Collaborative AR in STEM Classrooms by the group of authors Villanueva A., Zhu Z., Liu Z., Peppler K., Redick T., and Ramani K. and the article with the topic AR application as a digital experiment for education - An example in the Earth-Moon System by the group of authors Lindner C., Rienow A., Jürgens C. both with 39 citations.

Table 6. Top 10 Articles Ranked by Citation Count

CR	Title	Journal Title	TC	TC/ Year
1	Active learning augmented reality for STEAM education- a case study (Jesionkowska et al., 2020)	Education Sciences	74	24.67
2	Impact of augmented reality lessons on students' STEM interest (Hsu et al., 2017)	Research and Practice in Technology Enhanced Learning	73	12.17
3	Content delivery using augmented reality to enhance students' performance in a building design and assembly project (Shirazi & Behzadan, 2015)	Advances in Engineering Education	53	6.63
4	Enhancing 21st century skills with AR: Using the gradual immersion method to develop collaborative creativity (Sanabria & Arámburo-Lizárraga, 2017)	Eurasia Journal of Mathematics, Science and Technology Education	48	8.00
5	The Effect of augmented reality on students' learning performance in STEM education (Petrov & Atanasova, 2020)	Information (Switzerland)	46	15.33
6	Augmented reality enhanced cognitive engagement: designing classroom-based collaborative learning activities for young language learners (Wen, 2021)	Educational Technology Research and Development	44	22.00
7	Prospects of using the augmented reality application in STEM-based Mathematics teaching (Kramarenko et al., 2020)	CEUR Workshop Proceedings	42	14.00
8	A framework utilizing augmented reality to improve critical thinking ability and learning gain of the students in Physics (Faridi et al., 2021)	Computer Applications in Engineering Education	40	20.00
9	Meta-AR-App: An Authoring Platform for Collaborative Augmented Reality in STEM Classrooms (Villanueva et al., 2020)	Conference on Human Factors in Computing Systems - Proceedings	39	13.00
10	Augmented Reality applications as digital experiments for education – An example in the Earth-Moon system (Lindner et al., 2019)	Acta Astronautica	39	9.75

Notes: TC: Total citations; CR: Citation ranking

Main Research Topics

Using VOSviewer software, the study produced a network of keywords that appeared in at least two publications related to AR technology in STEM education. Each circular node represents a keyword, with its size reflecting how frequently it appears. The connections between nodes indicate co-occurrences in related studies, and keywords that are closely linked are depicted in the same color. In addition, associated keywords are drawn with the same color. Figure 6 shows that besides the two main keywords related to the research content, "AR" and "STEM education," there are also prominent keywords, "VR," "mobile application," "gamification," "chemistry," "higher education", "STEAM," "serious game.". As can be seen, the keywords are divided into two main groups representing the respective research directions.

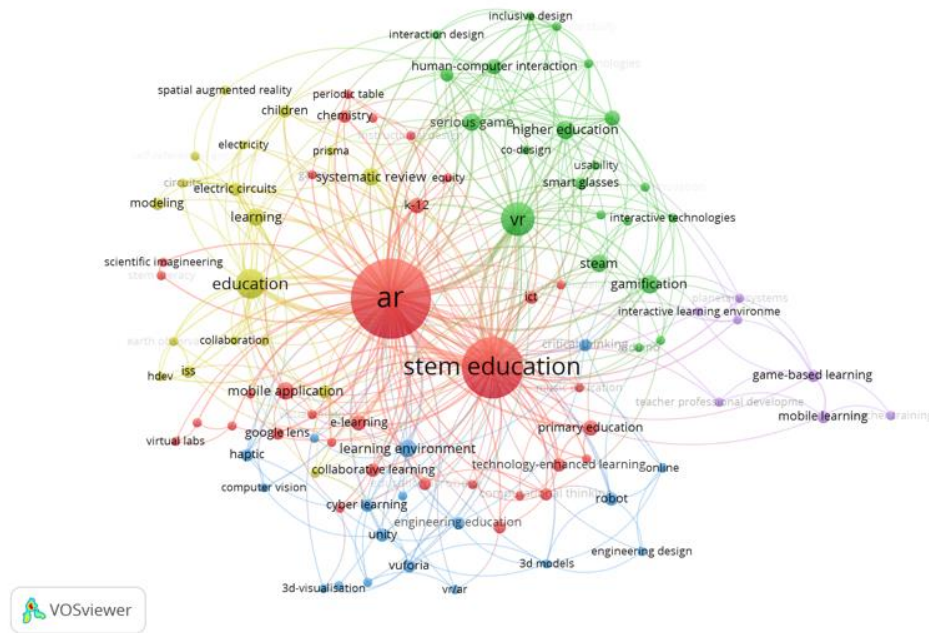


Figure 6. Network Visualization of 97 Keywords Co-Occurring in Two or More Publications (Source: Authors' own elaboration using VOSviewer)

The red cluster: Research on applying AR technology to STEM education (notably Chemistry), which combines learning forms of technology-enhanced learning, e-learning, and collaborative learning with other learning support tools and media such as mobile applications. Integrated AR technology onto mobile devices used in STEM education appears in many studies in the bibliography.

The green cluster: Applied studies combining AR and VR in STEM/STEAM education at the university level, with attention to gamification elements. There are also several other research directions, such as Studies on the application of AR/VR technology in changing the learning environment when training technical occupations (the blue cluster with the keywords robot, cyberlearning, machine learning); Research on applying AR technology in STEM education to teach physics concepts, spatial augmented reality, through AR/VR impact on the sense of touch (the yellow cluster); Research related to professional development and teacher training on using game-based learning in teaching with the support of AR/VR software on mobile (the purple cluster).

To further confirm the research direction, we analyzed Author keywords in the most recent 5-year collection period (2019-2023); the results are shown in Figure 7. The keywords in the publications are: "AR," "STEM Education," and "VR." These are the frequently used keywords in Figure 6 corresponding to the research trends in clusters 1 and 2. Research with the appearance of these keywords is from 2019 and 2020, in the most published articles in 2021. Research trends in the recent 2-year period (2022-2023) are with the keywords: "technology," "machine learning," "computational thinking," "e-learning," "robot," and "education technology." These keywords also appear in other research trends in Figure 6, proving that these are emerging research issues but are of interest.

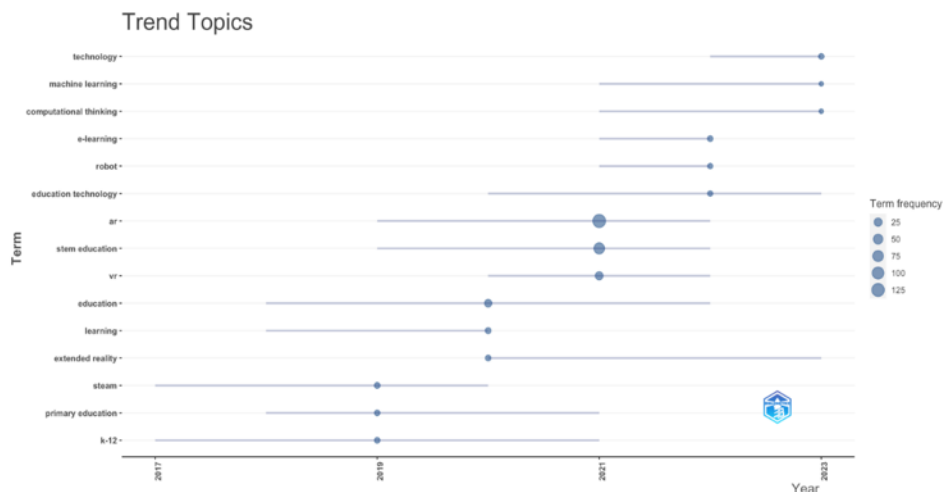


Figure 7. Trends in Author Keywords (Source: Authors' own elaboration using Biblioshiny), Period 2019 - 2023)

Figure 8 shows the trend of interest and development of the author keywords. In particular, we are especially interested in the keywords that appear in the pink and green areas. Keywords in the green area include "higher education," "serious games," and "chemistry." These are topics that have been of interest to authors recently. These keywords also appear in Figure 6 in the first research trend. Phrases appearing in the pink area are "AR," "STEM Education," and "VR," showing that these are topics of interest in the previous period but are not recent research trends.

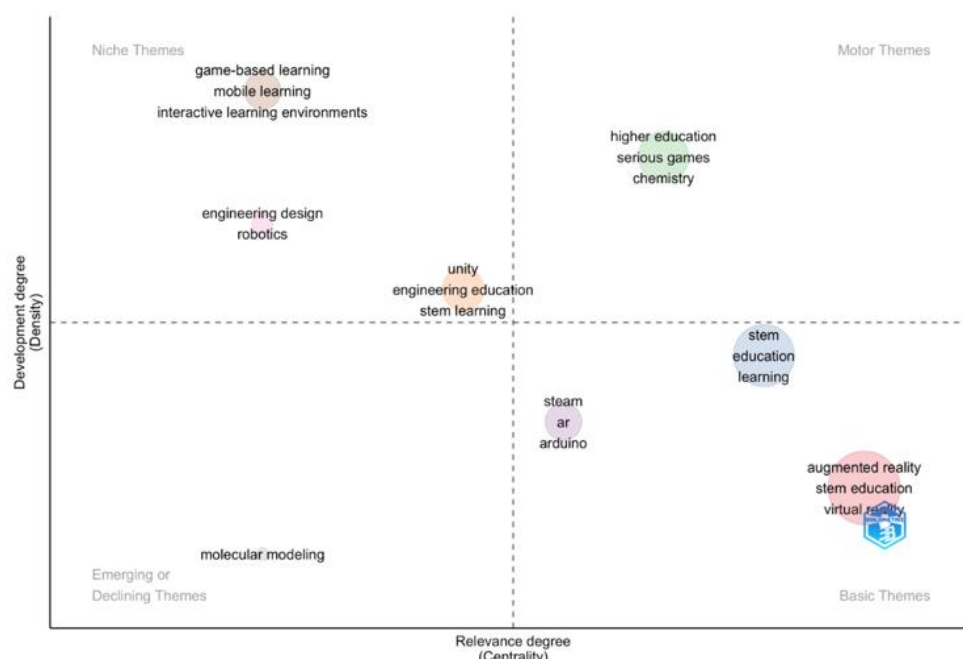


Figure 8. Thematic Representation of Research Directions in AR Applications for STEM Education (Source: Authors' own elaboration using Biblioshiny)

Conclusion

In this study, the authors examine global research trends concerning the use of AR technology for STEM education by analyzing data from Scopus with bibliometric methods. Key aspects of research on applying AR technology to STEM fields are also highlighted.

Bibliographic data from the Scopus database were utilized to create an overview and discuss publications related to the application of augmented reality technology in STEM education from 2005 to 2023. Research on AR's use in STEM education first appeared in Scopus in 2005; however, significant scientific findings from case studies in this field have emerged primarily over the last decade. The results on the number of publications and growth trends show that 2020-2022 was the period with the strongest increase in the number of articles and citations (see Figure 3). During the COVID-19 pandemic, virtual technology-based learning methods facilitated students' learning, as they could not attend school and had to rely heavily on virtual learning tools to meet pandemic prevention requirements (Reshi et al., 2023). AR products were evaluated as effective in supporting learning during this period (Putra et al., 2022). In Figure 3, the data on publications and citations for 2023 is incomplete because the data query time is December 2023, and many scientific publications in 2023 have not yet been updated to the Scopus database (Chadegani et al., 2013).

Analyzing the contributions of countries on this research topic, we found that among these ten countries, only two countries, the United Kingdom and Ukraine, have several citations/articles higher than the average of the entire bibliography (see Table 2). The United Kingdom has an outstanding number of citations because there is 01 article providing information about the current situation and research directions on the topic of virtual laboratories in science, engineering, and technology by the group of authors Potkonjak et al. (2016) with 523 citations (also the article with the most citations in the bibliography). This index shows that the quality of articles from this country is the highest in the bibliography. Of the total of 409 citations of 59 articles from the United States, only 1 article has an outstanding number of citations, contributing mainly to the country's total number of citations, "Content delivery using augmented reality to enhance students' performance in a building design and assembly project" by the group of authors, Shirazi A., Behzadan A.H. with 53 citations. Therefore, the TC/TP index of the United States is not high (6.93) despite contributing an outstanding number of articles in the bibliography (59 out of 201). In Ukraine, the TC/TP index is 15.33, showing that research by the authors from this country has a higher citation rate than the average of the bibliography. Besides, from the summary information in Table 5, Figure 4, and Figure 5, there is little international cooperation between authors and countries.

Although the countries Germany, Portugal, Brazil, and Ireland have institutions that appear in the ranking of the nine institutions contributing the most articles, research on this topic in these countries is not as consistent as it is in Ukraine and the United States, where each country has only one institution in the rankings (see Table 3). An interesting point is that the top 9 institutions out of 160 institutions in the bibliography have an average citation index of these institutions (TC/TP) of 13.95, not much higher than the average citation index of the entire bibliography (accounting for 13.28), proving that there is no big difference in the quality of articles from the institutions contributing the most articles compared to the remaining institutions.

Analyzing the results obtained regarding publication sources, authors, and most influential publications, we found that publications on this issue are mainly published in conference proceedings (see Table 4). This is consistent with the results of some previous studies, which have shown that research in the natural sciences tends to be published more in conference proceedings than in journals (Kochetkov et al., 2022). The Top 9 authors with the most publications on this issue are divided into three research groups, each group consisting of scientists from the same institution. In general, the publications from these three research groups have an average citation index (TC/TP) higher than the average of the entire bibliography (see Table 5). Based on the top 10 most influential articles of the bibliography, we see two distinct research directions: first, the effects and effectiveness of AR on enhancing learning performance and learning interest of students following the STEM education model; second, specific learning projects for students, which apply AR in STEM education. In research on the number of citations by year of the above articles, we found that all these articles had an insignificant number of citations before 2020. From 2020 to now, the number of citations has increased dramatically, showing that this research topic has only received attention in the last four years (see Table 6).

The keywords "AR," "STEM Education," and "VR" are also prominent in research applying AR technology within STEM education, as reflected in the WoS database (Talan, 2021; Zhao et al., 2023). Keywords such as "AR," "STEM Education," and "VR" show significant interest in integrating this technology into STEM education. This development opens up new opportunities in teaching methods, enhancing learning experiences, especially when combining AR with mobile applications and gamification (Ramli et al., 2024; Yegorina et al., 2021). The keyword analysis results show two main research directions in this field (Figure 6, Figure 7). First, integrating AR into collaborative learning and e-learning has improved students' problem-solving and visualization skills, especially in Chemistry (Abd Majid & Abd Majid, 2018; Lee & Tucker-Kellogg, 2020; Woźniak et al., 2020). Second, the combined use of AR and VR creates a highly interactive learning environment, allowing students to perform simulated experiments without equipment. This technology also supports special needs groups, such as children with autism (Sokolikj et al., 2023), and is effective in STEAM education, including music (Ho et al., 2023). In addition, the trend of integrating machine learning and robotics in STEM education is gradually growing, reflecting the growing demand for advanced and innovative learning methods in this field (Cecil et al., 2021; Iqbal & Campbell, 2023).

The emergence of keywords such as "machine learning," "computational thinking," and "robots" in recent studies indicates a new trend in integrating advanced technologies into STEM education. Studies in this period, such as those of Costa et al. (2021) and Iqbal and Campbell (2023), have shown that the combination of AR and these new technologies can bring significant improvements in STEM education, especially in addressing students' learning challenges and skill development.

This study thoroughly overviews AR technology's use in STEM education, emphasizing key elements such as the field's growth, leading contributors, and collaborative efforts. As the adoption of AR technology in STEM education rapidly expands, researchers frequently present case studies, making this evaluation valuable for monitoring progress in the area. Regular bibliometric reviews are crucial to keeping up with the evolving research landscape of AR applications in STEM education.

To promote a more balanced global development in the research on AR application in STEM education, it is essential to strengthen international collaboration, particularly between leading countries like the United States and others. Additionally, expanding research on AR application to other STEM fields beyond Chemistry, such as Mathematics, Physics, and Engineering, will help diversify the research scope and enhance inclusivity. Future studies should further explore the integration of AR with advanced technologies such as artificial intelligence (AI) and the Internet of Things (IoT) to maximize educational efficiency. Another area for improvement is the timeliness of the data, which is analyzed only up to 2023, while AR and STEM education are rapidly evolving fields.

This study only utilizes data from the Scopus database, which may not fully capture the global landscape of research on AR in STEM education. Integrating data from other sources, such as WoS or Google Scholar, would provide a more comprehensive view. Additionally, the study does not delve into factors influencing the development of this field, such as educational policies, technology availability, and the acceptance levels of teachers and students. Another limitation is the timeliness of the data, which is analyzed only up to 2023, while AR and STEM education are rapidly evolving fields.

Conflict of Interest

The authors disclose that they have no actual or perceived conflicts of interest. Artificial intelligence was not used in the preparation of this paper.

Fundings

This research is funded by the Thai Nguyen University of Education under grant number TNUE-2024-01.

Authorship Contribution Statement

Giang: Conceptualization, writing, original draft, editing, and visualization. Dinh: Writing, review & editing, formal analysis, and methodology. Ngu: Validation and supervision. Chau: Formal analysis, and methodology. N. P. Thao: Writing, original draft, writing, review & editing. T. T. P. Thao: Conceptualization, writing, original draft, writing, review & editing.

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